

LA RIOJA, Argentina

WMO: 872170

Lat: 29.382S

Lon: 66.793W

Elev: 429

StdP: 96.28

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.8	3.3	-4.9	2.6	8.8	-3.1	3.1	8.2	8.0	12.3	6.3	12.0	1.8	270	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	39.1	23.5	37.6	23.2	36.2	22.9	25.9	34.6	25.1	33.8	24.4	32.9	3.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.5	19.3	30.6	22.7	18.4	29.6	22.0	17.6	28.7	83.2	34.7	79.5	33.8	76.6	33.1	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.2	5.3	-1.4	42.3	1.8	1.3	-2.7	43.2	-3.7	44.0	-4.7	44.7	-6.0	45.6		
(5)			-2.7	27.7	1.8	1.3	-4.0	28.6	-5.1	29.4	-6.1	30.1	-7.4	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	27.9	26.7	24.4	20.3	16.2	12.6	12.1	15.4	19.3	23.3	25.7	27.5
	DBStd	6.64	3.18	3.52	3.47	3.66	3.54	2.95	3.12	3.98	4.34	4.20	3.80	3.44
	HDD10.0	31	0	0	0	1	2	9	14	4	1	0	0	0
	HDD18.3	636	0	1	2	20	87	172	196	109	39	8	1	0
	CDD10.0	4014	556	467	448	309	193	88	78	170	279	411	472	542
	CDD18.3	1577	298	234	192	79	19	1	1	17	68	161	223	284
	CDH23.3	17477	3534	2589	1727	592	137	11	32	269	736	1760	2564	3525
	CDH26.7	8821	1929	1329	764	194	23	1	7	94	293	854	1364	1967
Wind	WSAvg	2.3	2.4	2.2	2.2	1.9	1.8	1.7	2.0	2.3	2.5	2.7	2.7	2.8
Precipitation	PrecAvg	370	83	72	58	27	6	4	4	5	5	16	38	63
	PrecMax	640	228	203	205	193	27	32	27	54	29	73	195	218
	PrecMin	178	8	5	2	0	0	0	0	0	0	0	0	0
	PrecStd	117	50	50	47	30	7	7	6	11	6	19	31	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	39.7	37.1	34.0	29.0	24.8	26.7	33.8	36.0	38.9	40.1	41.2
		MCWB	25.3	24.4	24.0	23.0	20.1	17.0	16.1	18.8	19.5	21.9	22.6	23.1
	2%	DB	38.5	37.2	34.9	31.0	26.7	22.3	23.4	29.0	32.4	36.0	37.6	38.6
		MCWB	24.5	24.0	23.2	21.7	19.1	15.1	14.8	17.0	18.7	21.7	21.9	23.2
	5%	DB	36.7	35.4	33.0	28.8	24.7	20.8	21.4	26.1	30.0	33.7	35.7	37.0
		MCWB	23.9	23.7	22.7	20.6	17.9	14.3	13.7	15.7	17.6	20.4	21.6	22.9
	10%	DB	34.9	33.6	31.0	26.8	22.6	19.2	19.4	23.6	27.6	31.5	33.7	35.1
		MCWB	23.4	23.3	22.1	19.6	16.8	13.5	12.5	14.4	16.3	19.4	21.0	22.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	26.6	25.9	24.1	21.2	18.1	17.2	19.6	21.5	24.4	24.6	25.9
		MCDB	37.5	34.7	34.4	31.7	26.7	24.0	24.7	31.1	33.6	35.4	36.0	36.5
	2%	WB	26.1	25.6	24.6	22.8	19.9	16.2	15.6	17.7	19.8	22.4	23.5	24.8
		MCDB	34.8	33.5	32.0	28.7	25.2	21.0	22.3	27.9	30.6	33.4	34.6	35.3
	5%	WB	25.1	24.9	23.7	21.7	18.7	15.0	14.2	16.3	18.4	21.3	22.5	24.1
		MCDB	33.6	32.9	30.3	26.9	23.4	19.5	20.4	25.3	28.1	31.9	33.0	34.1
	10%	WB	24.3	24.1	22.8	20.7	17.4	13.9	13.0	14.8	17.0	20.1	21.7	23.2
		MCDB	32.5	31.7	29.1	25.5	21.7	18.2	18.5	22.8	26.1	29.5	31.6	32.7
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.5	10.6	10.8	10.7	12.3	13.8	14.4	13.9	13.1	13.0	12.9
		MCDBR	14.4	14.0	13.6	13.6	13.4	14.8	16.9	18.2	16.7	15.8	15.7	15.7
		MCWBR	5.2	5.2	5.2	6.5	7.3	8.8	9.8	9.4	7.9	7.0	6.4	5.9
	5% WB	MCDBR	12.4	11.9	11.4	11.5	11.6	13.3	15.3	17.0	15.4	14.1	14.0	13.6
		MCWBR	5.3	5.0	4.7	5.9	6.6	8.2	9.3	9.1	7.7	7.0	6.3	5.7
Clear-Sky Solar Irradiance	taub		0.396	0.385	0.369	0.353	0.326	0.308	0.302	0.335	0.377	0.376	0.380	0.391
	taud		2.418	2.454	2.495	2.492	2.511	2.529	2.522	2.411	2.302	2.366	2.403	2.413
	Ebn at Noon		946	938	921	886	864	860	883	889	896	936	955	953
	Edh at Noon		125	117	106	96	85	79	82	102	126	126	126	126
All-Sky Solar Radiation	RadAvg		7.28	6.51	5.55	4.48	3.55	3.30	3.68	4.67	5.83	6.85	7.57	7.65
	RadStd		0.38	0.44	0.37	0.36	0.33	0.24	0.19	0.24	0.36	0.43	0.38	0.41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+0.85	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page